INIXA PROMPT BATTLE ARENA

Comprehensive College Event Rulebook, Game Specs & Host Playbook

1. Executive Overview & Competition Goals

The **INIXA Prompt Engineering Battle** is a premier competitive gaming format designed for college inter-symposiums, technical fests, and AI hackathons. Unlike standard programming contests that test code syntax, the Prompt Battle evaluates participants' **mental model of Large Language Models (LLMs)**, prompt optimization techniques, constraint satisfaction, and precision AI control.

- **Equal Playing Field:** All contestants utilize the INIXA AI Gateway backend, ensuring zero model bias or API key advantage.
- **Pure Technical Focus:** 100% focused on Text Research, Algorithmic Optimization, and Schema Transpilation (zero image/video distraction).
- **Audience Entertainment:** The Finals feature a 1v1 Auditorium Stage Projector view with real-time token stream & live score dials.

2. Detailed Specifications of All 6 Competition Games

Game 1: Unstructured Data to Typed JSON (Text Research)

Category: Text Research | **Difficulty:** Medium | **Time Limit:** 4 Minutes

Goal: Contestants receive a noisy support call transcript. They must write a prompt instructing the LLM to extract entity metadata into strict valid JSON format.

Sample Input Context:

```
Transcript: Customer John Doe (ID 9842) called complaining that desktop app on Windows 11 crashes when uploading > 50MB files. Agent found buffer overflow in upload streaming service. Recommending patch v2.4.1 deployment by EOD.
```

Target Goal Output:

```
{ "userId": 9842, "customerName": "John Doe", "os": "Windows 11", "severity": "HIGH", "rootCause": "Buffer overflow in upload streaming service", "recommendedFix": "Deploy patch v2.4.1" }
```

✓ Constraint 1: Output MUST be 100% valid JSON parseable by JSON.parse().

✓ Constraint 2: Do NOT wrap in markdown fences (```json ... ```).

✓ Constraint 3: Zero conversational fluff (No 'Here is your JSON:').

Game 2: Zero-Hallucination Academic Extraction (Text Research)

Category: Text Research | **Difficulty:** Hard | **Time Limit:** 5 Minutes

Goal: Extract exact quantitative statistics from a dense academic study summary into a GitHub Markdown table. The prompt must enforce strict zero-hallucination guardrails.

Target Goal Output:

```
| Cohort | Sample Size (n) | Retention Increase (%) | Dropout Rate (%) | P-Value | |---|---|---|---|---| | Group A | 120 | 34.2% | 5% | < 0.001 | | Group B | 115 | 4.1% | 12% | < 0.001 |
```

✓ Constraint 1: Output MUST be a GitHub Markdown table only.

✓ Constraint 2: Strict zero-hallucination rule (no estimated numbers).

✓ Constraint 3: Token cap under 50 tokens total.

Game 3: Constrained Roleplay & Reasoning (Text Research)

Category: Text Research | **Difficulty:** Extreme | **Time Limit:** 3 Minutes

Goal: Prompt the AI model to explain Quantum Entanglement to a high schooler under extreme negative constraints.

✓ Constraint 1: ■■ **EXTREME RULE:** Do NOT use the letter 'e' anywhere in the output!
✓ Constraint 2: Must explain entanglement accurately in under 60 words.
✓ Constraint 3: Maintain an encouraging mentor persona.

Game 4: Algorithmic Optimization O(N^2) -> O(N) (Coding Battle)

Category: Coding Battle | **Difficulty:** Medium | **Time Limit:** 5 Minutes

Goal: Refactor a slow nested-loop duplicate detector function into an optimal O(N) HashMap solution in TypeScript.

Sample Input vs Target Goal:

```
// Input: Slow nested loop O(N^2) function // Target: Optimal Set/Map TypeScript implementation achieving strictly O(N) runtime with JSDoc documentation.
```

- ✓ Constraint 1: Time complexity must be strictly O(N).
- ✓ Constraint 2: Must include complete JSDoc typed docstrings.
- ✓ Constraint 3: Must be pure TypeScript with zero npm external packages.

Game 5: Reverse Code Engineering (Coding Battle)

Category: Coding Battle | **Difficulty:** Hard | **Time Limit:** 5 Minutes

Goal: Contestants receive input data `[1, [2, [3, 4], 5], 6]` and expected flat output `[1, 2, 3, 4, 5, 6]`. Prompt the AI to write a clean recursive flattening function.

- ✓ Constraint 1: Do NOT use built-in Array.prototype.flat() or flatMap().
- ✓ Constraint 2: Must handle arbitrary array nesting depth.
- ✓ Constraint 3: Include clear inline comments explaining recursion stack.

Game 6: Zod & React Hook Form Schema Transpiler (Coding Battle)

Category: Coding Battle | **Difficulty:** Extreme | **Time Limit:** 6 Minutes

Goal: Transpile OpenAPI backend specifications into complete Zod validation schemas with inferred TypeScript types in one prompt.

- ✓ Constraint 1: Must export both `z.object({...})` schema AND `type UserRegistration = z.infer`.
- ✓ Constraint 2: Include custom validation error messages.
- ✓ Constraint 3: 100% production ready TypeScript code.

3. Prompt Engineering Master Cheatsheet (For Participants)

Participants should apply these core prompt engineering strategies to achieve top scores:

Technique	Implementation Pattern
Role Assignment	*'You are a Lead Data Architect. Your task is to...'* - Sets persona, tone, and domain expertise.
Delimiters	Use triple quotes **"""**, XML tags **<context>...</context>** to isolate input data from instructions.
Negative Guardrails	Explicitly state *'DO NOT include intro text, DO NOT wrap in markdown fences'*.
Few-Shot Examples	Provide 1 or 2 sample input-output pairs inside the prompt for complex schema matching.
Chain-of-Thought (CoT)	Prompt the model to *'Think step-by-step before producing the final JSON'* for logic puzzles.

4. Supported AI Models in INIXA Arena

Contestants can select from these flagship AI models during competition rounds:

Model Name	Engine Type	Best Used For
GPT-5.6 (UPDF Flagship)	UPDF Stream API	Complex multi-step reasoning, JSON schema extraction & zero-shot tasks.
DeepSeek V4 Pro	G4F Dedicated	Deep algorithmic coding, TypeScript refactoring, and logical puzzles.
Qwen 3.7 Max (Worker)	Alibaba Worker	High speed instruction following, markdown table formatting & strict constraint matching.
Qwen 3.7 Plus (Worker)	Alibaba Worker	Ultra-low latency streaming for high-speed speed run rounds.
Perplexity Copilot	Direct Worker	Factual verification and academic data extraction.
GPT-5.4 Mini	Surfsense API	Lightweight, high-speed execution for short constrained tasks.

5. Auditorium Stage 1v1 Setup & Official Judging Rubric

Host Playbook for Stage Finals:

1. Open the INIXA App on the stage host computer connected to the auditorium projector.
2. Click on the **Stage Projector (1v1)** toggle in the top-right header.
3. Player 1 (Left Screen) and Player 2 (Right Screen) laptops connect to stream live prompt typing.
4. The projector displays live split screen showing live prompt drafts, generated AI outputs, and real-time score dials.
5. The Automated AI Judge computes scores immediately upon submission, announcing the round winner!

Weighted Scoring Rubric:

Evaluation Metric	Weight	Evaluation Standard
Target Output Accuracy	**40%**	Degree to which generated output matches target JSON/code/table specifications.
Constraint Adherence	**30%**	100% compliance with negative rules (e.g. no letter 'e', no markdown fences, O(N)).
Token & Speed Efficiency	**20%**	Concise prompt construction with minimal token waste and fast completion time.
Technique & Elegance	**10%**	Use of delimiters, role assignment, and few-shot/CoT prompt engineering techniques.

INIXA AI Studio — Designed for College Prompt Engineering Battles & Hackathons